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Phytoremediation of Cd-Contaminated Soil Assisted by Plant Growth Promoting Rhizobacteria: Comparative Review

Abstract: Heavy metal contamination in soil is toxic and harmful to organisms and human health. As an eco-friendly solution to removing cadmium (Cd) from the soil, phytoremediation is effective on its own, but it has limitations. This comparative study reviews an innovative microorganism-plant approach to improving the efficiency of phytoremediation in the case of Cd-contaminated soils. The authors of selected papers use a hyperaccumulator plant (sorghum and dandelion, respectively) and rhizobacteria strains with known plant growth-promoting features (PGPR) in their well-designed and controlled pot experiments. Within a relatively short period (3-4 months), both studies demonstrate the success of PGPR in enhancing the phytoextraction of Cd content from soil, working well with complex soil contamination situations (Cd+PE) and other organic compounds such as tea saponin. Through careful procedures of the harvested plant biomass, soil properties analysis, spectroscopy or spectrometry for Cd, and DNA sequencing, the data and results reveal that PGPR can achieve such enhancement by altering the bacterial communities, improving the soil nutrition, and increasing the enzyme activities in the rhizosphere. In conclusion, PGPR-assisted phytoremediation is a more efficient strategy than traditional plant-only phytoremediation for reducing Cd in the soil.

Keywords: Phytoremediation; PGPR; Rhizosphere

1. Introduction

Soil contamination is one of the major environmental issues in our modern society today. Cadmium (Cd) pollution in soil poses a dangerous threat to organisms and human health. Having a feasible and sustainable strategy to remove toxic heavy metals from soil is an important field since many countries are working hard to achieve the Sustainable Development Goals set out by the UN, such as “sustainable cities and communities”, “clean water and sanitation”, and “sustainable use of terrestrial ecosystems”. In order to solve this problem, phytoremediation is often adopted as the eco-friendly and cost-effective strategy for Cd-contaminated soil [1]. Many studies have been conducted in the literature on hyperaccumulators and their capacities for heavy metal uptake. The results are promising. For instance, a pot of castor-oil plant (10.5g dry weight) can accumulate up to 155.1 µg of Cd after just 2 months of growth. [2]

While phytoremediation is considered an environmentally friendly and cheaper solution to contaminated soils, the traditional phytoremediation strategies and processes are often slow and require a long duration. Moreover, when the soil is contaminated with a high concentration (the bioavailable portion) of heavy metal, most plants' growth is damaged or suppressed. Thus, their potential for remediation is limited [3]. So, it is worth the effort to review some of the novel phytoremediation strategies to see how we can improve their efficiency. One approach is to utilise plant growth-promoting rhizobacteria (PGPR), which have been tested to show great potential in enhancing the oilseed rape plant's capacity to remove the organic pollutant of phenanthrene (PHE) from soils [3].

In this comparative review, two studies (Liu et al. 2024 and Yu et al. 2023) are selected for their new methods in which they seek out help from microorganisms and/or organic compounds to enhance the phytoremediation results in Cd-contaminated soils [4-5]. The authors use PGPR in their controlled treatments and then analyse the data extracted from the plant biomass and soil after a relatively short-term pot experiment to verify the long-term potential of the PGPR in cadmium phytoremediation strategies.

2. Results and Discussion

2.1 PGPR enhance phytoextraction of Cd from soil

Both studies unequivocally demonstrate the success of PGPR in enhancing the efficiency of phytoremediation of Cd-polluted soils.

Under the PGPR treatments, the accumulation of Cd content by the sorghum plant in Study 1 is drastically increased (more than 50% and up to 96.9%); Cd is accumulated mainly in the aboveground parts with higher Cd concentration in the belowground parts [4]. Similarly, Yu et al. (2023) in Study 2 also find a significant increase in total Cd extraction when adding PGPR (without tea saponin) to the treatments, with a higher Cd concentration in shoots ($0.15 \text{ mg} \cdot \text{kg}^{-1}$ in shoots vs $4.21 \text{ mg} \cdot \text{kg}^{-1}$ in roots) [5].

Both studies' results show that the residual fraction of Cd in the soil is subsequently reduced by adding PGPR to the phytoextraction experiment. Furthermore, Study 2 analyses the bioconcentration factor (BCF) and the transport factor (TF) of the hyperaccumulator, suggesting a plant defence mechanism is in place to protect itself from heavy metal stress. The addition of PGPR enhances BCF in roots and amplifies the plant's TF, thus promoting Cd solubility and mobilisation in the soil [5].

The data about the plants' roots (Cd content, BCF, and TF) align with another finding from these two studies: PGPR promote growing much longer roots (up to 33.3% increase in sorghum [4], 53.2% in dandelion [5]). This explains a key aspect of phytoremediation – the Cd accumulation and distribution happen in the rhizosphere, or simply put, near the roots. A more developed root system may play an important role in the success of a phytoremediation strategy.

2.2 PGPR work with complex soil contamination situations and treatments

Liu et al. (2024), in Study 1, find that the presence of microplastic (PE) in the soil worsens the negative effect of Cd in sorghum plant growth (both biomass and growth rate). However, thanks to the inoculation with two PGPR strains, the toxicity is alleviated, and the tested plants grow much more biomass under these inoculated treatments [4].

Yu et al. (2023), in Study 2, explore an additional treatment of the soil: tea saponin (Ts), which changes the concentration of bioavailable Cd in the soil even before the inoculation of the PGPR [5]. Although another assisting material is used, the results of Study 2 still show that PGPR are the overpowering factor in enhancing the decontamination effects of the hyperaccumulator plant. The PGPR-only treatment group have higher (better) results than the Ts-only treatment group in terms of plant growth, root length, and enzyme activities analysis [5].

With the combination of PGPR and Ts in the treatment, results are much better compared to those of using a single-assistance (e.g. only Ts or only PGPR) treatment [5]. This implies that PGPR work well with another compatible enhancer. Identifying organic compounds with specific properties that interact with the heavy metals in soil can potentially enhance phytoremediation efficiency even further.

2.3 How PGPR get the job done

The selected papers study and reveal how PGPR improve the phytoremediation efficiency in the context of Cd contamination – through 1) altering the bacterial

communities, 2) improving the soil nutrition (nutrient bioavailability), and 3) enhancing the enzyme activities in the rhizosphere.

Basically, the PGPR treatments increase the bacterial diversity in the rhizosphere, as shown in Study 1, the higher Shannon index, Simpson index, ACE index, Chao1 index, and Sobs index [4]. Through DNA sequencing, Study 2 finds that a few Cd-tolerant bacteria, such as *Actinobacteria* and *Gemmatimonadota*, are relatively more abundant in the microbial community composition after inoculating PGPR – this significantly helps the hyperaccumulator plant to tolerate Cd and thus purify Cd in the soil much better [5].

In both studies, results show that PGPR alter the adverse effects of Cd on soil nutrition and create a more friendly environment for the hyperaccumulator plants – decreasing (slightly) the pH value in soil and increasing the potassium content [4] and total nitrogen and phosphorus [4,5].

Additionally, PGPR enhance the antioxidant enzyme activities, as shown in Study 2, through higher catalase (CAT) and peroxidase (POD) activities in dandelion roots [5].

The positive effects of PGPR in all of the areas mentioned above contribute to the overall improvement in removing the heavy metal Cd from the soil, transforming it into a less harmful form for the environment and the plant within a relatively short period of time.

3. Materials and Methods

Both studies focus on testing a PGPR-assisted method of improving phytoremediation efficiency in cadmium-contaminated soils. Experimental structures are similar: controlled pot experiment, plant biomass and soil sample preparation and analysis, determination of Cd content, and soil microbial organisms DNA sequencing analysis. However, in detail, they differ in terms of actual materials used (plants, soils, bacteria, and chemical compounds) and the specific procedures carried out based on those materials (Table 1).

Table 1. Comparison of key components from the two studies.

	Study 1 (Liu et al., 2024)	Study 2 (Yu et al., 2023)
Plant used:	Sorghum (<i>Sorghum bicolor</i>)	Dandelion (<i>Taraxacum platyepidum</i>)
Soil pollutant(s):	- Cd: 10 mg·kg⁻¹ (manually added) - PE: 0.05 & 0.25 mg·kg⁻¹ (manually added)	Cd: 3.47±0.12 mg·kg⁻¹ (present in sourced soil)
PGPR strain(s):	<i>Bacillus</i> sp. SL-413 (SUB13086935) <i>Enterobacter</i> sp. VY-1 (SUB13968556)	<i>Burkholderia cenocepacia</i> (NZ_JVDJ00000000.1)
Inoculation:	30, 45, 60 days post germination	Not specified
Other assisting materials:	None	Tea saponin (Ts); 1% mass fraction; added on 105th day.
Moisture level during cultivation:	75%	30%
Harvest time: (after seeding)	3 months (90 days)	4 months
Controlled experiment treatments:	8 groups: clean soil, Cd-contaminated soil, Cd+PE*-contaminated soils, Cd+PE*-contaminated soils with 2 PGPR strains.	5 groups: Cd-contaminated soil without dandelion, with dandelion, with Ts, with PGPR, with Ts+PGPR.

Notes: * PE concentration at two different levels.

Similarities:

When analysing the harvested plant samples, Study 1 considers both the above-ground and below-ground biomass to study the effects of two pollutants (Cd & PE) in the soil and two PGPR strains' effects [4]. Similarly, Study 2 conducts its data analysis around shoots and roots [5].

When determining the Cd content, Study 1 uses Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES). Similarly, Study 2 uses Graphite Furnace Atomic Absorption Spectrometry (GFAAS).

When analysing the bacterial communities in the rhizosphere, both studies apply the same DNA processing steps to the soil samples: DNA extraction, and then polymerase chain reaction (PCR) amplification using a standardised purification kit, and finally, sequencing on the Illumina MiSeq platform.

Key Differences:

As Table 1 shows, Study 1 not only looks into the heavy metal (Cd) contamination in soil but also investigates the effects of microplastics (PE), which should not be neglected when addressing today's soil problems. This Cd+PE complex contamination is studied through 8 carefully designed experimental groupings, including the control group, which consists of uncontaminated soil [4].

Study 2 uses a niche wild plant, dandelion, which is different from what Study 1 uses - sorghum, a commercial crop. They explore and widen the choices and potential of phytoremediation as similar studies in this field use a commercial crop or energy crop (e.g. rapeseed, castor-oil plant) [1-3].

The method in Study 2 is relatively new and innovative: they use an organic chemical compound (a surfactant) and living microorganisms as a combined strategy, and their experiment results have turned out to be promising [5].

4. Conclusion

The comparative review of the selected studies implies that when adopting a phytoremediation strategy, it is advisable to avoid relying solely on the hyperaccumulator plants to do the work. In the case of decontaminating Cd in soil, adding PGPR to the treatment is a proven productive enhancement choice because PGPR can alleviate the damage of soil pollutants (e.g. Cd, PE) to the plant; promote the plant to grow more biomass, especially roots, to accumulate Cd; improve soil nutrition, microbial diversity, and enzyme activities in rhizosphere; and overall enhance the phytoextraction of Cd in soil. Plus, PGPR can work well with another organic chemical compound, tea saponin, which boosts the results significantly. In conclusion, PGPR-assisted phytoremediation is more efficient than the traditional plant-only approach for soil cadmium pollution. There is potential in exploring such a mixed microorganism-biosurfactant phytoremediation strategy.

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